

General framework / things to think about when starting a new project

1. Problem Definition

- What exactly is the task? Is it well-defined?
- Who is the user? What do they actually need?
- What does success look like to them vs. to us?

2. Prior Work & Existing Solutions

- Has this been done before? What did they do?
- Are there existing models, datasets, or pipelines we can reuse?
- Are we in a research setting (novel contribution needed) or industry (best solution wins)?
- What are the limitations of prior work we need to address?

3. Data

- Does training/evaluation data exist?
- Can we use it? (licensing, copyright, consent, privacy)
- What does it look like? (domain, language, format, noise)
- How much do we have? Is it enough?
- If not — how do we get more? (annotation, synthetic data, data augmentation, scraping)
- Is the data representative of our real users?
- Are there biases in the data we need to worry about?

4. Constraints

- How much compute do we have?
- Should we use open or closed models?
- How big a model can we actually use? (memory, latency, cost)
- Are there latency requirements? (real-time vs. batch)
- On-premise or cloud? (data residency, privacy regulations)
- How long do we have to build this?

- How many people are working on it?

5. Architecture & Techniques

- What's the right approach? (prompting, fine-tuning, RAG, agents...)
- Do we need retrieval? A knowledge base? External tools?
- Do we need to train or fine-tune, or can we use off-the-shelf?
- How do we handle context length limitations?
- What are the tradeoffs of our architectural choices?

6. Evaluation

- What metrics do we use? (automatic vs. human)
- Do the metrics actually capture what the user cares about?
- What's our baseline?
- How do we know if we're improving?
- How do we evaluate things that are hard to measure? (helpfulness, tone, safety)
- Do we have a test set that reflects real usage?

7. Reliability & Safety

- Can the model hallucinate? How badly does that matter here?
- How do we detect and mitigate hallucinations?
- Is the model robust to noisy or unexpected inputs?
- What happens when it fails? Gracefully?
- Is there a human in the loop? Should there be?

8. Privacy & Legal

- Does the system handle sensitive data? (medical, legal)
- What regulations apply? (GDPR, HIPAA, etc.)
- Can we send data to an external API, or must it stay on-premise?
- Who owns the outputs? Any IP concerns?

- Are there terms of service issues with the models or data we're using?

9. Fairness & Bias

- Does the system perform equally well across user groups?
- Are there populations that could be harmed?
- How do we test for this?

10. Interpretability

- Do we need to explain the model's decisions? (to users? regulators? doctors?)
- What level of interpretability is required?
- How do we provide it?

11. Deployment

- How do we serve this at scale?
- How do we monitor it in production?
- What happens when the model is wrong in production?
- How do we update it over time? (continual learning, retraining, prompt updates)
- How do we version and document it?

12. Impact

- What's the potential positive impact?
- What could go wrong at scale?
- Who benefits and who might be harmed?